data collection protocol

Purpose

Define the minimum and complete set of raw inputs required to compute all signals, mathematical formulas, and algorithms in the system.

This document ensures:

every variable used in the system is traceable to a concrete, collectable data field.

1. Core Principle

All computations in the system must originate from:

User Event Logs

Each event must be recorded in a structured format.

2. Unified Event Schema

Every interaction generates one event:

{ “event_id”: “uuid”, “user_id”: “string”, “post_id”: “string”, “event_type”: “vote | post | view | share”, “vote”: -1 | 0 | +1, “timestamp”: “unix_ms”, “location”: { “lat”: float, “lon”: float, “accuracy”: float, “source”: “gps | wifi | ip | unknown” }, “device”: { “device_id”: “hashed_id”, “user_agent”: “string”, “os”: “string”, “browser”: “string” }, “network”: { “ip”: “string”, “ip_geo”: { “lat”: float, “lon”: float, “country”: “string” } }, “content”: { “text”: “string (only for post events)” } }

3. Mapping: Raw Data → Signals

3.1 Location-Based Signals

Required Fields

location.lat location.lon timestamp

Used In

- GPS consistency
- speed plausibility
- movement continuity
- location inconsistency
- proximity

Collection Method

Frontend: navigator.geolocation API Backend fallback: IP-based geo lookup

Demo Simplification

If GPS unavailable: use ip_geo only

3.2 Device Fingerprint

Required Fields

device.device_id device.user_agent device.os device.browser

Used In

- device consistency score

Collection Method

Generate device_id = hash(user_agent + screen + random_seed)

Demo Simplification

Use only user_agent hash

3.3 Network / IP Data

Required Fields

network.ip network.ip_geo

Used In

- IP consistency
- location cross-validation

Collection Method

Backend: extract IP from request use GeoIP database

3.4 Temporal Data

Required Fields

timestamp (high precision)

Used In

- rate deviation
- entropy
- timing irregularity
- session continuity

Requirement

millisecond precision preferred

3.5 Interaction Data

Required Fields

event_type vote post_id user_id

Used In

- credibility
- anomaly
- graph construction

3.6 Content Data

Required Fields

content.text

Used In

- urgency (keywords)
- category classification
- ML credibility

Demo Simplification

Use keyword dictionary instead of ML

4. Derived Data (Computed, Not Collected)

4.1 Ground Truth y_j

For Demo

y_j = majority_vote_after_threshold

Rule

if N_j > threshold AND Var_j < limit: y_j = sign(S_j)

4.2 Human Baselines

For Demo

Use fixed constants:

mu_human_session = 10 minutes sigma_human_session = 5 minutes v_max = 50 m/s

4.3 Keyword Scores

Define Dictionary

{ “fire”: 1.0, “accident”: 0.9, “urgent”: 0.8, “help”: 0.7 }

4.4 Category Score

Demo Rule

if keyword_score > threshold: Cat_j = 1 else: Cat_j = 0.3

4.5 Similarity Function

Define

Sim(j,k) = cosine(TF-IDF(j), TF-IDF(k))

5. Minimum Viable Data (For Demo)

If you want to simplify system:

Collect Only

user_id post_id vote timestamp text ip

Optional

location (if available) device_id (basic)

6. Data Flow Pipeline

User Action ↓ Frontend collects: location + device + action ↓ Backend enriches: IP + geo ↓ Store in Event Log DB ↓ Signal Computation Layer ↓ Mathematical Engine ↓ Propagation + Alerts

7. Storage Recommendation

Primary DB

PostgreSQL (structured data)

Cache

Redis (real-time signals)

Optional

Vector DB (for similarity)

8. Critical Constraints

Consistency

user_id must be stable across sessions

Privacy (Important)

hash device_id do not store raw sensitive data